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Choroidal melanoma

Long-term outcomes of eye-conserving treatment with Ruthenium[106] brachytherapy for choroidal melanoma ☆

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ABSTRACT

Purpose: To evaluate long-term outcomes of eye-conserving treatment using Ruthenium-106 plaque brachytherapy with or without transpupillary thermotherapy (TTT) for small to intermediate size choroidal melanomas.
Methods: Outcomes of 425 consecutive patients were analysed. The median basal tumour diameter was 10.9 mm (range 4.8–15.9 mm), and the median apical height 4.2 mm (range 1.2–9.3 mm). Brachytherapy doses ranged from 400 to 600 Gy with TTT (86%), or from 600 to 800 Gy without TTT (14%), specified at the scleral surface. Kaplan–Meier survival curves, log-rank tests and Cox regression analysis were used for analysis.
Results: Median follow-up was 50 months. Five-year actuarial local control was 96%. Five-year overall and metastases-free survival rates were 79.6% and 76.5%. Prognostic factors for metastasis-free survival were peripheral location ($p = 0.02$) and smaller basal diameter ($p < 0.001$). No dose effect relationships were found. Radiation side effects were frequent, with 2- and 5-year rates free of radiation complications of 60% and 35%. Five-year enucleation rate was 4.4% (10 for local recurrence, 7 for complications). Cosmetic and functional (visual acuity >0.10) eye preservation rates were 96% and 52% at 5 years.
Conclusions: Ruthenium-106 brachytherapy for choroidal melanoma provides excellent rates of local control and eye preservation.

Choroidal melanoma is the most common primary eye cancer in adults. Until the introduction of plaque brachytherapy in the 1960s, enucleation was standard treatment. From that time on, various eye-conserving treatment modalities such as Ruthenium-106 (Ru-106) or Iodine-125 plaque brachytherapy, proton beam radiotherapy, stereotactic radiotherapy, transscleral or transretinal local resection, and phototherapy (photocoagulation or transpupillary thermotherapy, TTT) have been developed with the aim of preserving useful vision without increasing the risk of metastatic spread [1]. Several studies have demonstrated that eye-conserving treatment with plaque brachytherapy does not increase the risk of distant metastases [2,3]. The Collaborative Ocular Melanoma Study (COMS) confirmed the efficacy of plaque brachytherapy in a multicenter randomised trial including 1317 patients, which did not show any difference between enucleation and I-125 brachytherapy in 5-year survival (81% vs. 82 %) or 5-year rates of death with melanoma metastases (11% vs. 9%) [4].

Although the most important treatment aim for patients with choroidal melanoma is survival free of disease recurrence, both patients and physicians focus on the preservation of useful vision, cosmetic appearance and quality of life as well. Since its introduction, Ru-106 brachytherapy has been increasingly used for treatment of small to medium sized choroidal melanomas [5–7]. Although the benefit of eye preservation may be reduced by impaired vision resulting from radiation toxicity, eye conservation has major functional and cosmetic advantages.

Previously, our group has reported outcome data after Ru-106 brachytherapy for choroidal melanoma [8,9], as well as results of the combination of brachytherapy and transpupillary thermotherapy (TTT) [10,11]. This combination therapy (so-called sandwich therapy) was initiated to use lower radiation doses for smaller tumours to reduce the risk of severe radiation toxicities, and on the other hand enable eye-conserving treatment for patients with tumours thicker than 5 mm. Moreover, patients with insufficient tu-

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